

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 1. Use Case Diagram

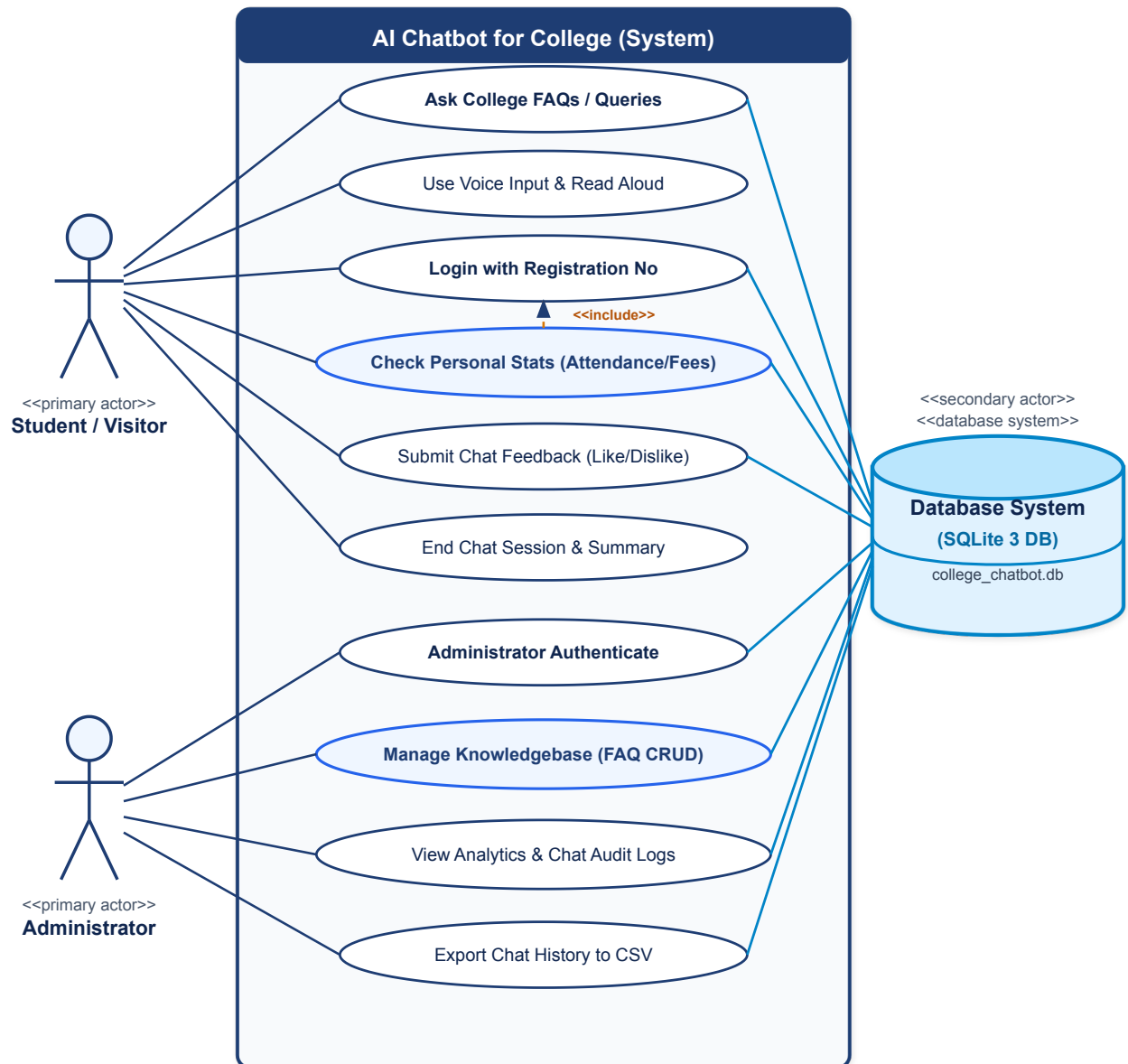


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 2. Class Diagram (With Full Relations & Multiplicities)

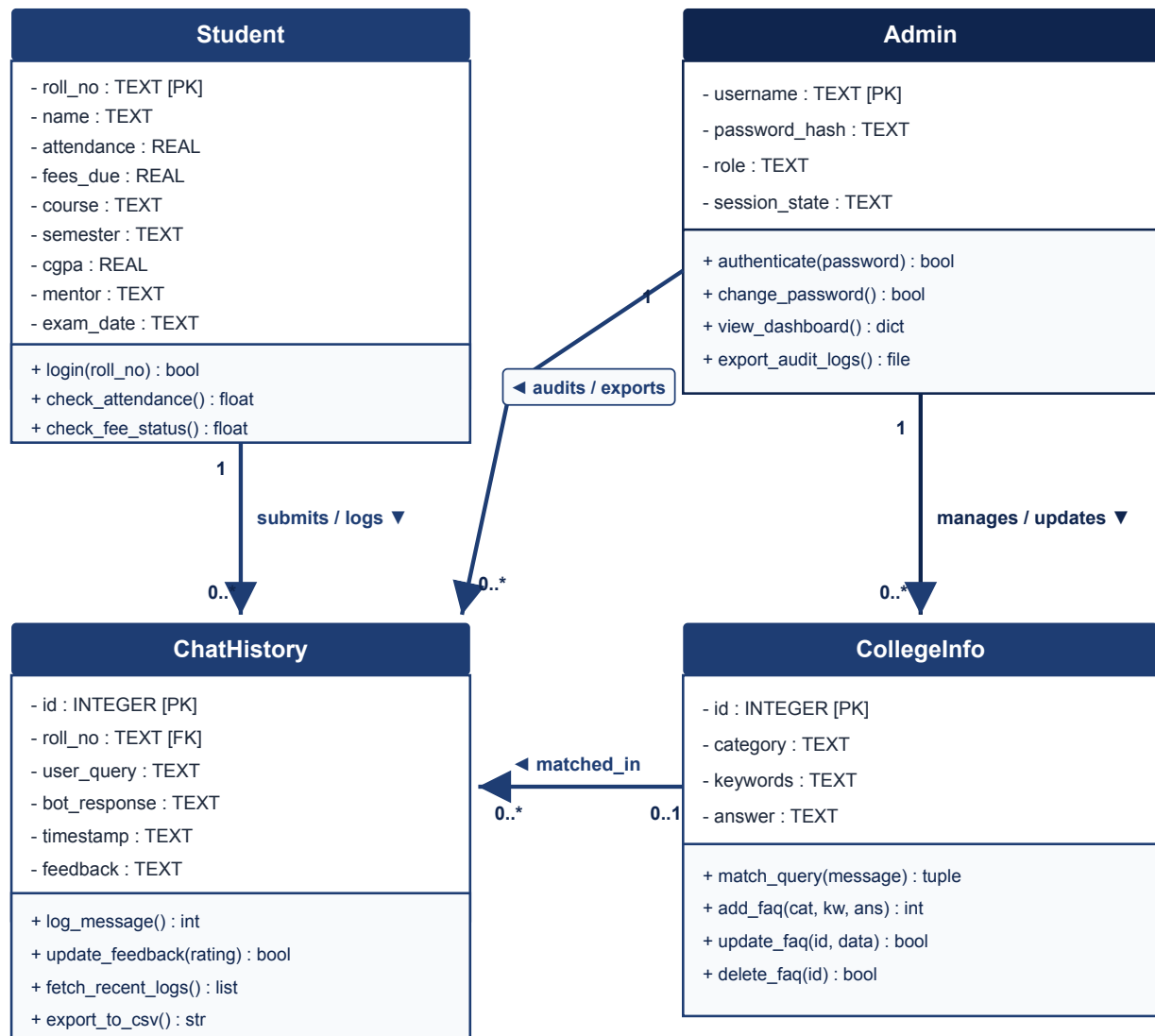


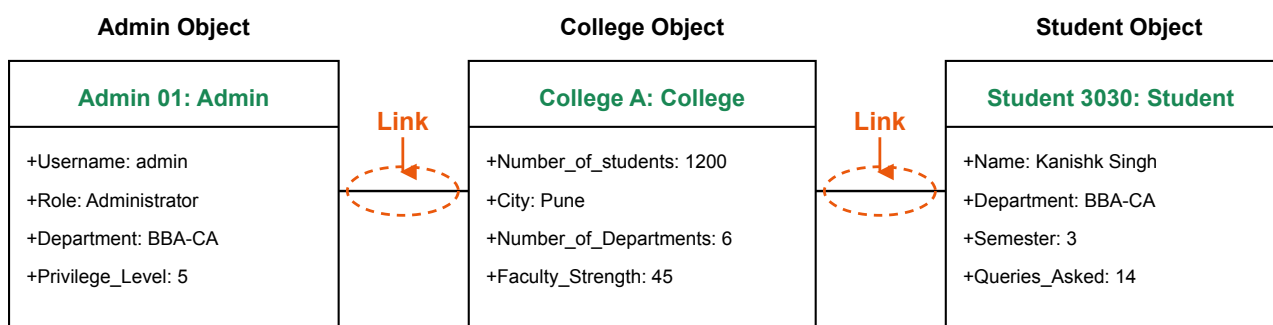
Fig 2: Class Diagram — core entities, attributes, operations, and 100% solid association arrows with multiplicities and role verbs.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 3. Object Diagram (Runtime Instances & Slot Values)

An object diagram using a link and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Object Diagrams | Unified Modeling Language (UML)

GfG

Fig 3: Object Diagram — runtime object instances (Admin Object, College Object, Student Object) connected via formal UML links matching GeeksforGeeks standard notation.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 4. Sequence Diagram — Query, Feedback & Session Termination

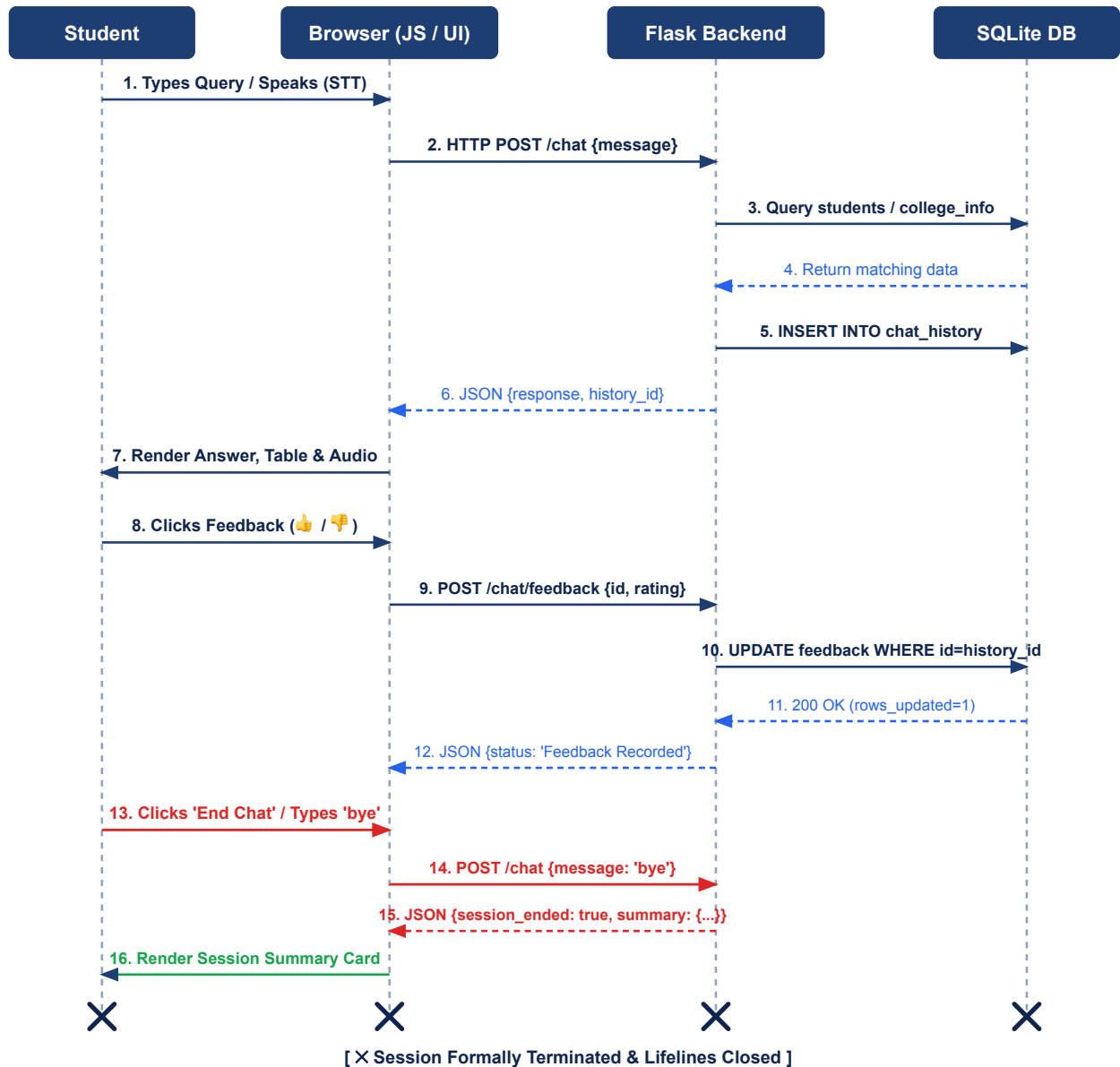


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (X).

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 5. Activity Diagram — Chatbot Response & Session Flow

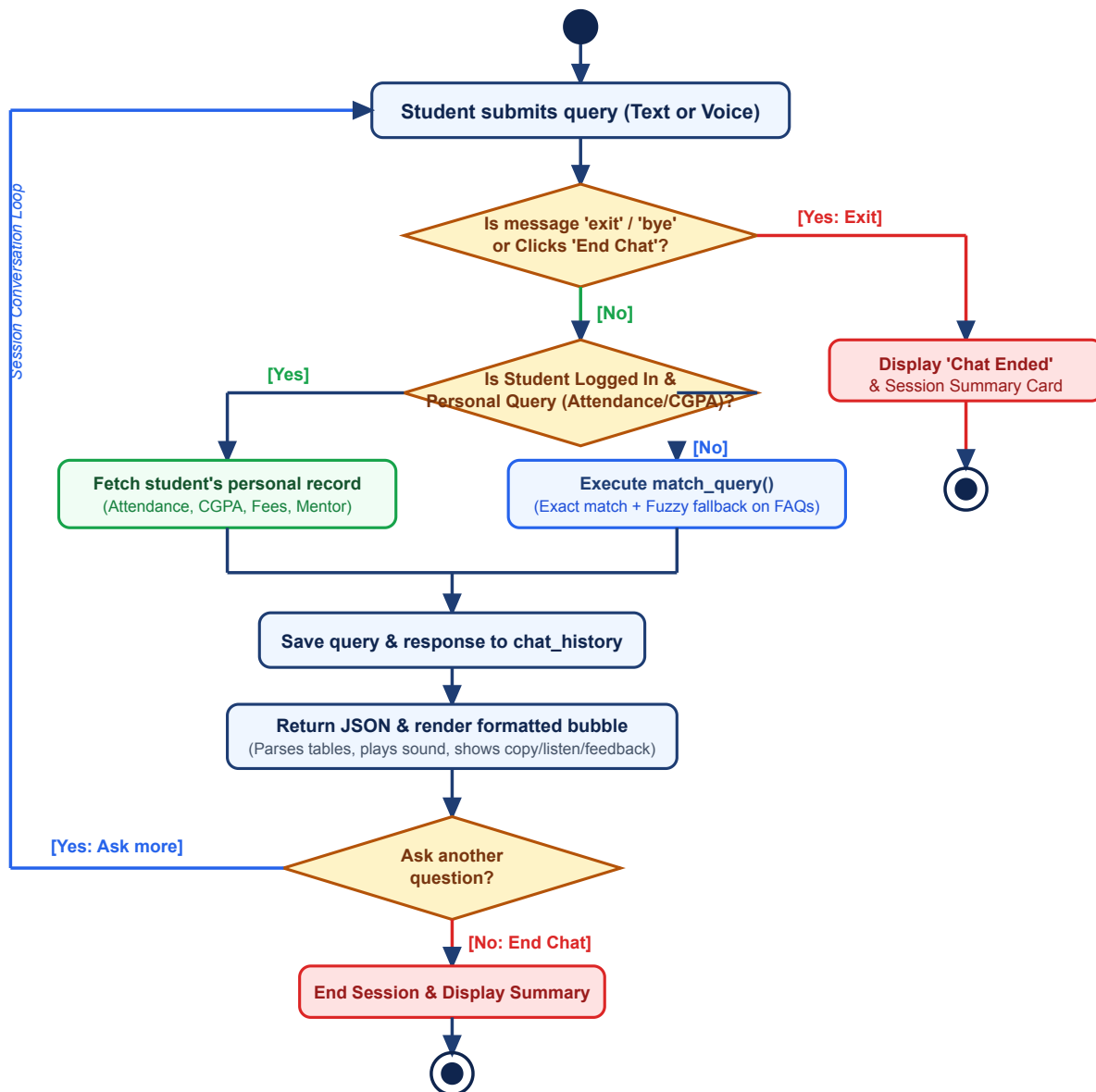


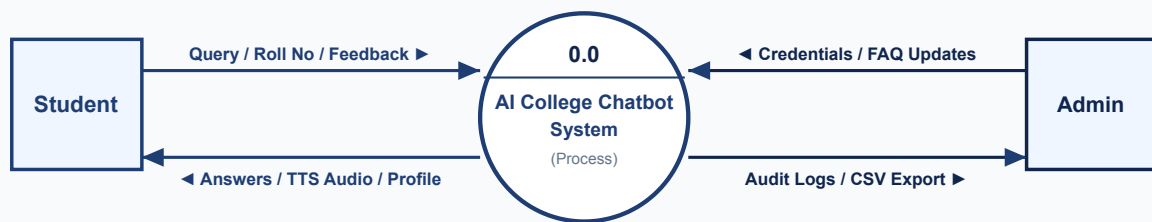
Fig 5: Activity Diagram — decision flow from receiving a message to displaying the response, showing the controlled session loop and explicit End state.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)

DFD Level 0: Context Diagram



DFD Level 1: Process Decomposition

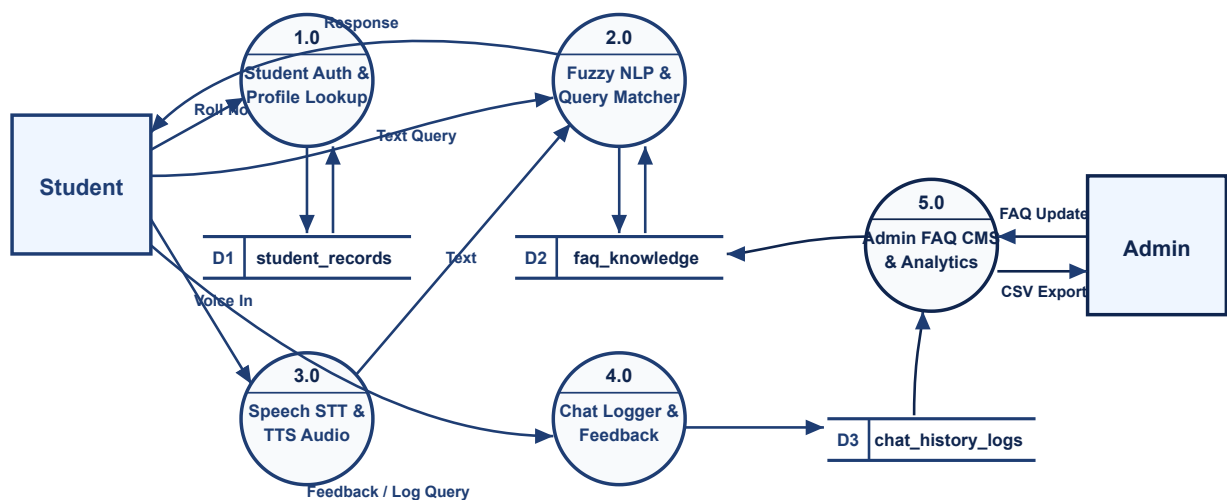


Fig 6: Data Flow Diagram (DFD) — Level 0 Context & Level 1 Process Breakdown using square entity symbols, circle process symbols, and solid data flow arrows.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 7. Entity-Relationship Diagram (ERD — Database Architecture)

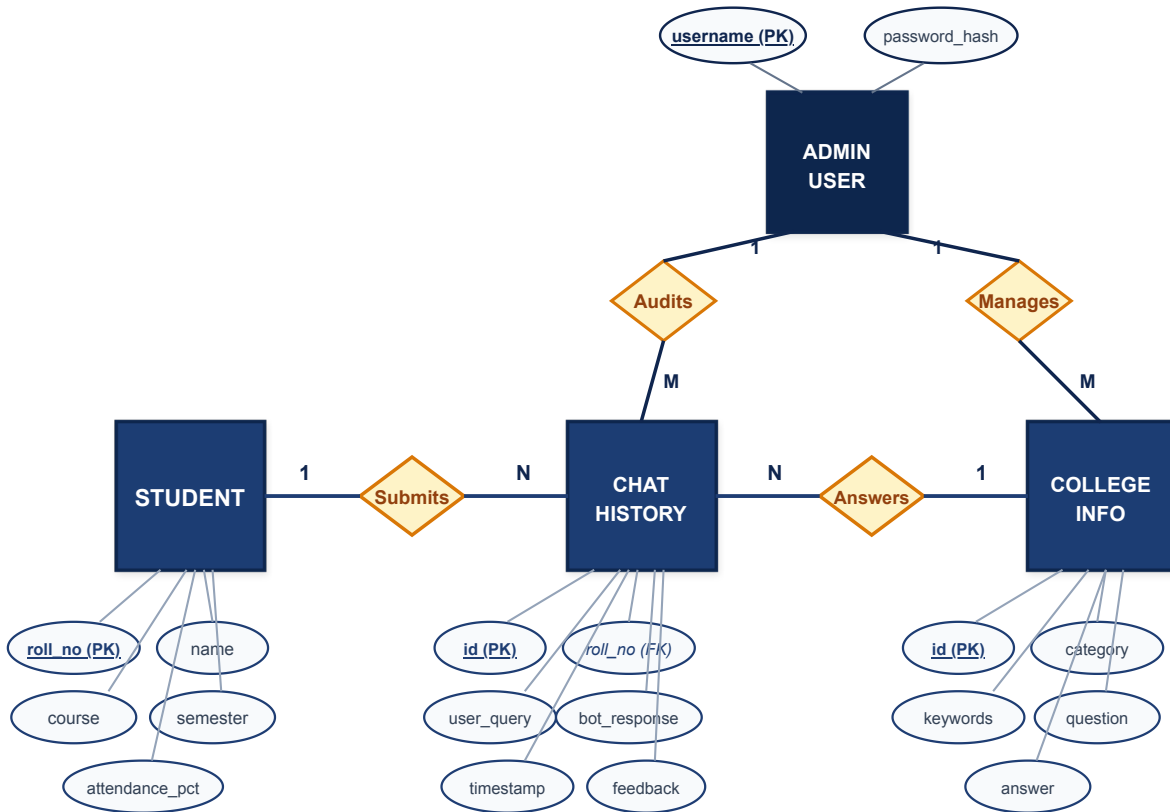


Fig 7: Entity-Relationship Diagram (ERD) — structured relational flow using square entity symbols (STUDENT, ADMIN\_USER, CHAT\_HISTORY, COLLEGE\_INFO) with cardinalities and non-crossing attributes.

# SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

## 8. Deployment Diagram (Hosting Topology & Artifact Architecture)

deployment Development Environment Model

### Development Environment Model

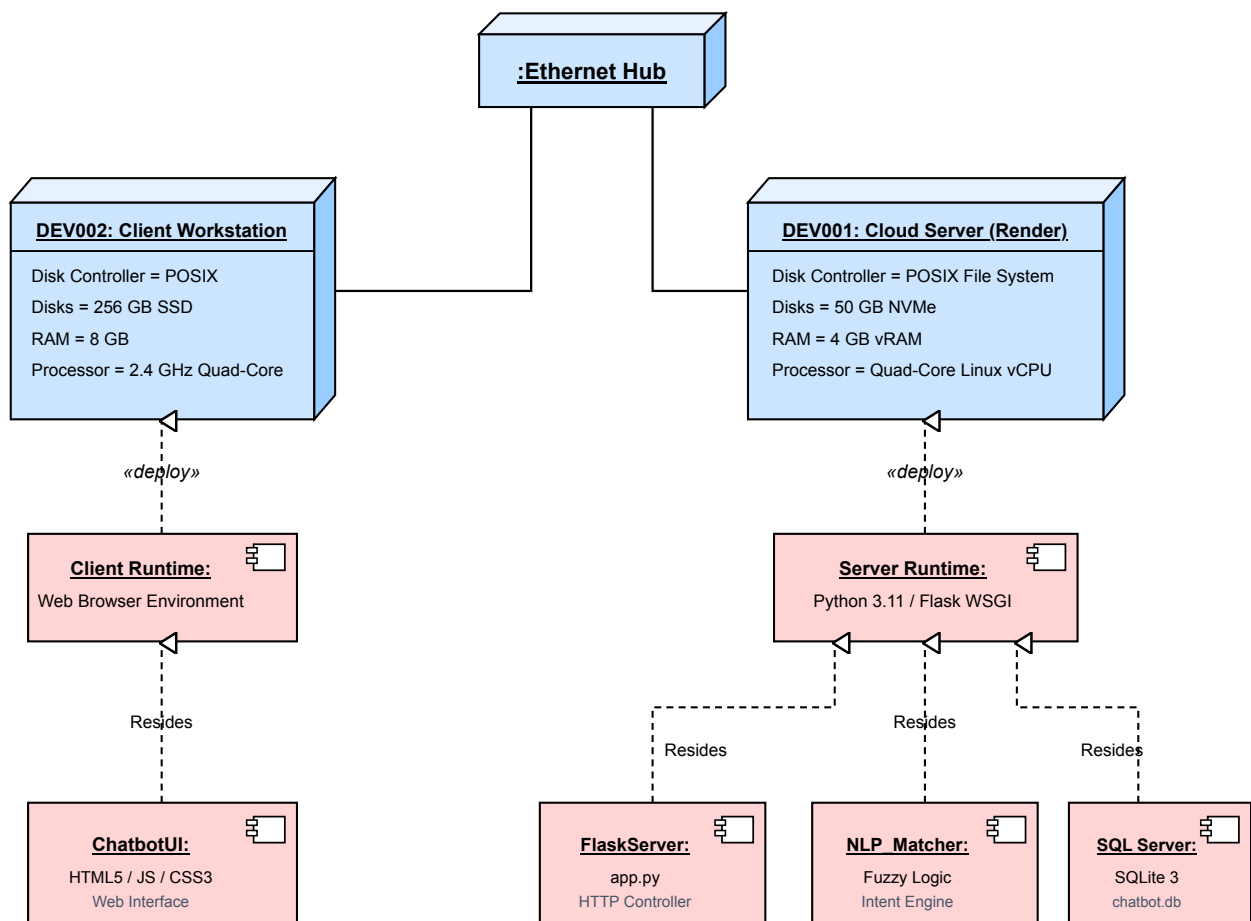


Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.